



Assignment 02

Introduction to software Engineering

Professor:

Engr. Muhammad Umer Haroon

Educator at FAST-NUCES | C/C++

Web Developer

Submitted by:

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Muhamad

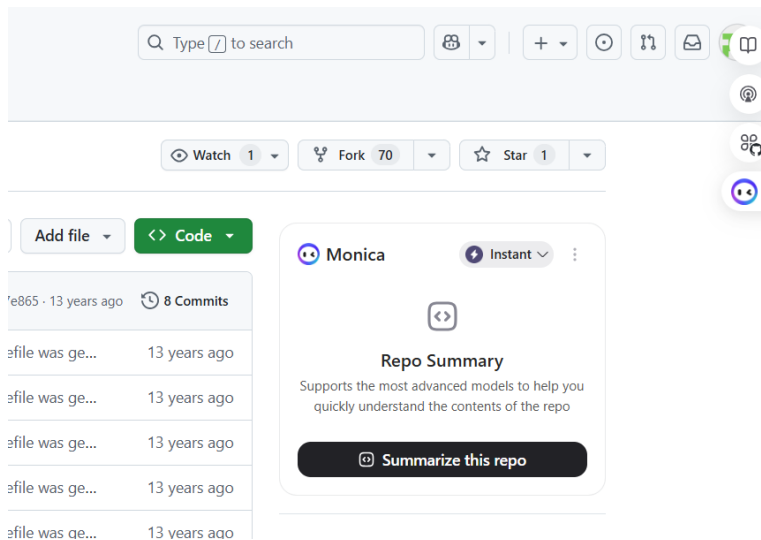
Muzammil Ayan

1. Project Overview

This project refactors a legacy-style C++ Loan/EMI calculator. Key objectives included: Fixing bugs and adding input validation Removing hardcoded values and supporting configuration files Refactoring code into multiple files Introducing long double to prevent overflow Adding unit tests Documenting using Doxygen

2. Bug Fixes & Refactorings Major bug fixes were applied, including: Fix 1: Input validation for principal, interest, and tenure (Change by Javed Ali) Fix 2: Preventing integer overflow by using long double (Change by Javed Ali) Fix 3: Replaced hardcoded magic numbers with config.txt (Change by M. Muzammil) Refactoring included splitting main.cpp into: Loan.cpp Loan.h Utils.cpp Utils.h

Step 01 : forked the repository



Step 02: git clone the repository on local pc

```

MINGW64~/c/Users/Admin/legacy-calc-2009
Admin@DESKTOP-QI6H2EA MINGW64 ~
$ git clone http://github.com/Javed640/LoanCalculatorCpp
fatal: destination path 'LoanCalculatorCpp' already exists and is not an empty d
irectory.

Admin@DESKTOP-QI6H2EA MINGW64 ~
$ cd legacy-calc-2009

Admin@DESKTOP-QI6H2EA MINGW64 ~/legacy-calc-2009 (main)
$ git add .

Admin@DESKTOP-QI6H2EA MINGW64 ~/legacy-calc-2009 (main)
$ git -m "fixed loancalculator zero-interest calculation bug"
unknown option: -m
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
      [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
      [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy
      [--no-optional-locks] [--no-advice] [--bare] [--git-dir=<path>]
      [--work-tree=<path>] [--namespace=<name>] [--config-env=<name>=<envva
      r>]
      <command> [<args>]

Admin@DESKTOP-QI6H2EA MINGW64 ~/legacy-calc-2009 (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:   https://github.com/Javed640/LoanCalculatorCpp/pull/new/main
remote:

```

Task 01

Building and running in console mode:

```
g++ -std=c++11 *.cpp -o calc
```

```
./calc -cp -a 100000 -i 12 -n 360
```

```
Loan #1:
Principal: 100000.00
Annual Rate (%): 12.00
Years: 30
Is Valid: Yes
Simple total: 3360000.00
Monthly (simple): 9333.33
EMI: 1028.61
-----
Loan #2:
Principal: 50000.00
Annual Rate (%): 7.25
Years: 5
Is Valid: Yes
Simple total: 268125.00
Monthly (simple): 4468.75
EMI: 995.97
-----
Loan #3:
Principal: 250000.00
Annual Rate (%): 4.20
Years: 30
Is Valid: Yes
Simple total: 7815000.00
Monthly (simple): 21708.33
EMI: 1222.54
-----
Loan #4:
Principal: -1000.00
Annual Rate (%): 5.00
Years: 2
Is Valid: No, NEGATIVE/ZERO input detected
```

Add a .gitignore file for build artifacts (e.g., *.o, calc, *.exe)

Commits

master

Commits on Nov 25, 2025

Update .gitignore with complete build artifact rules

Javed640 committed 9 hours ago

Commits on Apr 18, 2012

Added compilation information

bradyallenjohnson committed on Apr 18, 2012

LoanCalculatorCpp / .gitignore

Javed640 Update .gitignore with complete build artifact rules 2710713 · 1 hour ago History	
Code	Blame 15 lines (15 loc) · 109 Bytes
Older 14 years ago first version 1 2 3 4 5 <pre> 1 *. [sw] 2 *~ 3 *.o 4 build/ 5 noproject/ </pre>	
13 years ago Added a Qt GUI and some comp... 6 7 8 9 <pre> 6 -xcode [gs, -d311t+ 7 loanCalculator.exe 8 mac*.cc </pre>	
1 hour ago Javed640 Update .gitignore with complete... 10 11 12 <pre> 9 *.o 10 *.exe 11 calc </pre>	

Task 01

Task 3

Unit TEST

Normal EMI calculation

Invalid input handling

Large tenure calculations without overflow

```

MINGW64/d/calculator
Admin@DESKTOP-QI6H2EA MINGW64 /d/calculator/build
$ cmake --build . --target MyLoanTests
[ 28%] Built target gtest
[ 57%] Built target gtest_main
[ 71%] Building CXX object CMakeFiles/MyLoanTests.dir/tests/test_LoanCalculator.cpp.obj
[ 85%] Building CXX object CMakeFiles/MyLoanTests.dir/src/LoanCalculator.cpp.obj
[100%] Linking CXX executable MyLoanTests.exe
[100%] Built target MyLoanTests

Admin@DESKTOP-QI6H2EA MINGW64 /d/calculator/build
$ ctest --output-on-failure
Test project D:/calculator/build
Start 1: LoanCalculatorTests
1/1 Test #1: LoanCalculatorTests ..... Passed    0.31 sec

100% tests passed, 0 tests failed out of 1

Total Test time (real) = 0.35 sec

Admin@DESKTOP-QI6H2EA MINGW64 /d/calculator/build
$ ^C

Admin@DESKTOP-QI6H2EA MINGW64 /d/calculator/build
$ ./MyLoanTests --gtest_filter=* --gtest_color=yes --gtest_output=console
Running main() from D:/calculator/lib/googletest/googletest/src/gtest_main.cc
[WARNING] D:/calculator/lib/googletest/googletest/src/gtest.cc:5756:: WARNING: unrecognized output format "console" ignored.

[=====] Running 3 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 3 tests from LoanCalculatorTest
[ RUN      ] LoanCalculatorTest.NormalEMICalculation (0 ms)
[ RUN      ] LoanCalculatorTest.InvalidInputHandling (0 ms)
[ RUN      ] LoanCalculatorTest.InvalidInputHandling (0 ms)
[ RUN      ] LoanCalculatorTest.LargeTenureCalculation (0 ms)
[ RUN      ] LoanCalculatorTest.LargeTenureCalculation (0 ms)
[-----] 3 tests from LoanCalculatorTest (16 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 1 test suite ran. (32 ms total)
[ PASSED ] 3 tests.

Admin@DESKTOP-QI6H2EA MINGW64 /d/calculator/build
  
```

All 3 test cases are passed

Added doxygen commit as well as generated doxygen file

build	11/26/2025 3:04 AM	File folder
docs	11/26/2025 4:54 AM	File folder
html	11/26/2025 3:27 AM	File folder
☐ include	11/26/2025 2:42 AM	File folder
latex	11/26/2025 3:27 AM	File folder
lib	11/26/2025 2:41 AM	File folder
src	11/26/2025 2:42 AM	File folder
tests	11/26/2025 2:43 AM	File folder
CMakeLists	11/26/2025 3:04 AM	Text Document
Doxyfile	11/26/2025 3:26 AM	File

Task 01